PROJECT REPORT

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Branch: MCA **Section/Group: 24MCA2(B)**
Semester: 4th **Date of Performance: 21/04/2026**
Subject Name: Internet Of Things Lab **Subject Code: 24CAP-751**

Aim: RGB Color Detection and Visualization using Arduino and Processing Software

Objectives:

1. To detect different colors using an RGB color sensor.
2. To measure and display Red, Green, and Blue values.
3. To visualize detected colors in real-time using Processing software.
4. To understand interfacing between sensors and Arduino.

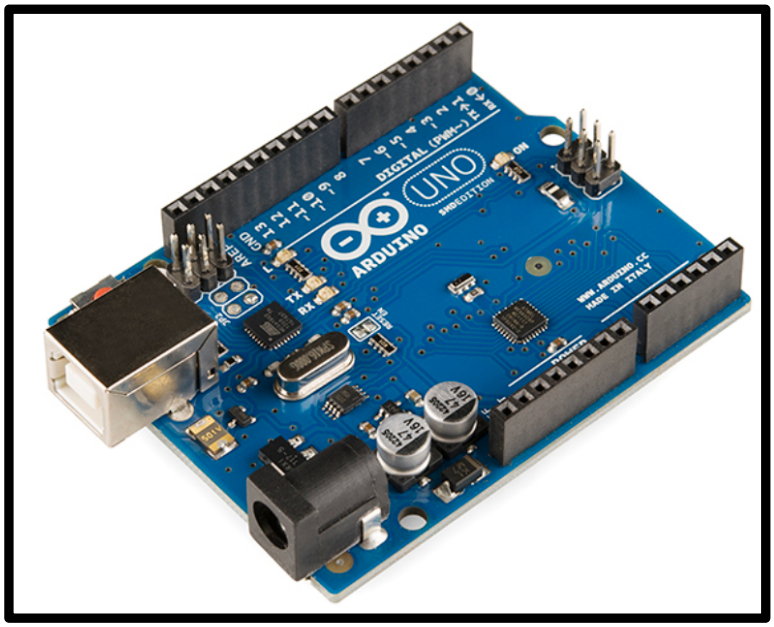
Components Required:

S.No	Component Name	Quantity
1	Arduino UNO	1
2	RGB Color Sensor (TCS3200/TCS34725)	1
3	RGB LED	1
4	Resistors (220Ω)	3
5	Breadboard	1
6	Jumper Wires	As required
7	USB Cable	1
8	Computer with Processing Software	1

Details of Components:

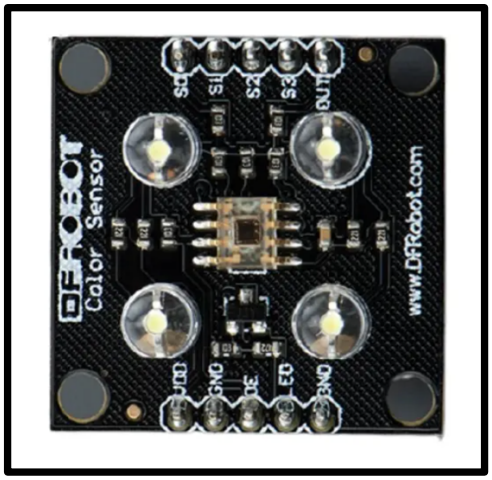
1. Arduino UNO:

Arduino UNO is a microcontroller board based on ATmega328P. It is used to read data from the color sensor and process it. It also sends data to the Processing software.



2. RGB Color Sensor:

The RGB color sensor detects the intensity of Red, Green, and Blue components of light. It converts the detected color into electrical signals which are read by Arduino.



3. RGB LED:

An RGB LED is used to display colors by combining red, green, and blue light in different intensities.

Block Diagram of Designed Model:

- The color sensor detects the color of the object
- It sends RGB values to the Arduino
- Arduino processes and sends the data through serial communication
- Processing software receives the data
- The detected color is displayed on the screen

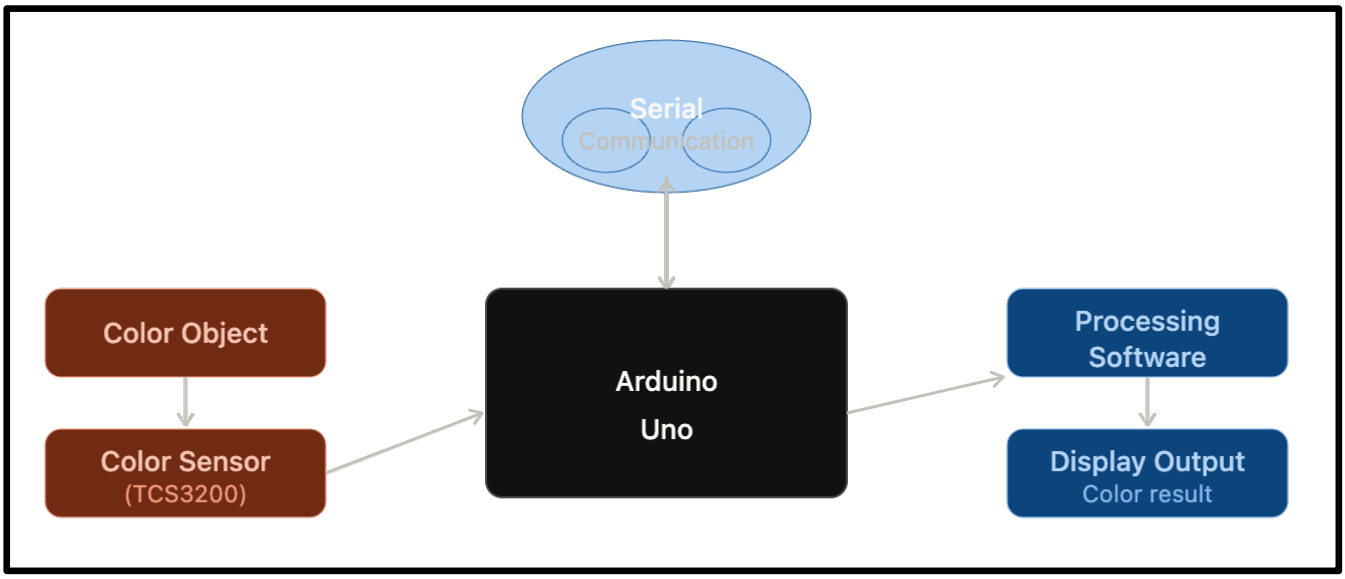


Figure 1 : Block Diagram of Model

Working of Designed Model:

- As illustrated in Figure X, when a colored object is placed in front of the color sensor, the sensor detects the intensity of Red, Green, and Blue (RGB) components present in the reflected light. These values represent the color characteristics of the object in digital form.
- The detected RGB values are then transmitted to the Arduino UNO, which acts as the central processing unit of the system. The Arduino reads these analog/digital signals and converts them into usable numerical data corresponding to each color component.
- The Arduino UNO processes the incoming RGB values and sends the data to the computer system using serial communication. This communication is established through the USB interface between Arduino and the system running the Processing software.
- The Processing software receives the serial data in real time and interprets the RGB values to generate a visual output. Based on the received values, it dynamically changes the background color of the display window to match the detected color.
- In addition to visual representation, the Processing interface also displays the exact numerical values of Red, Green, and Blue components. This allows precise monitoring and verification of the detected color.
- The system further analyzes the RGB values to determine the dominant color and displays the corresponding color name (such as Red, Green, or Blue) on the screen.

- For example, when the sensor detects values such as Red = 241, Green = 35, and Blue = 33, the system identifies Red as the dominant component. Hence, the detected color is displayed as **RED** on the output screen.
- Overall, the system performs real-time color detection and visualization by integrating hardware (sensor and Arduino) with software (Processing), ensuring accurate and immediate feedback.

Pictures of Prototype:

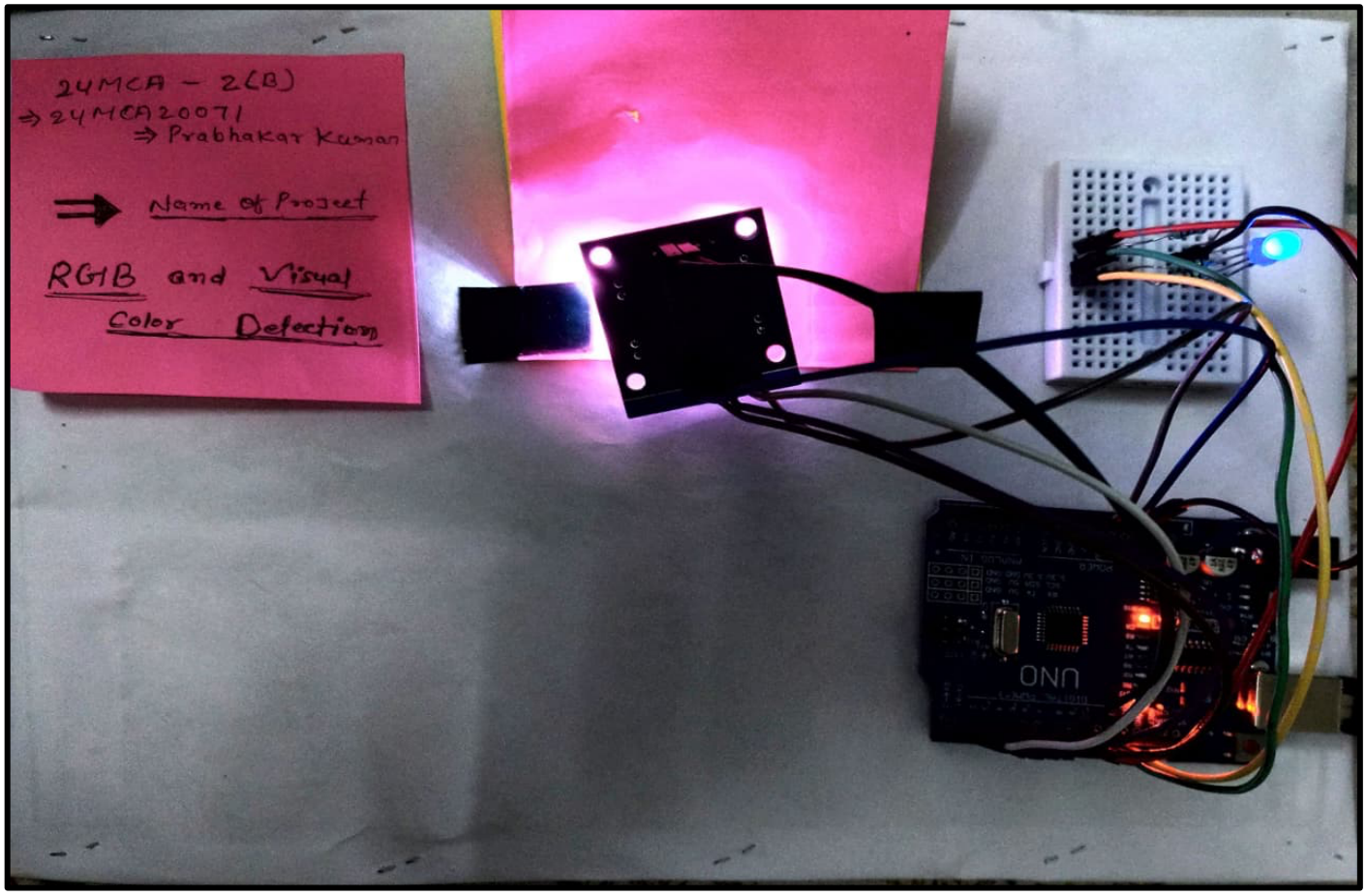


Figure 2 : Designed Model

Output of Deigned Model/Prototype:

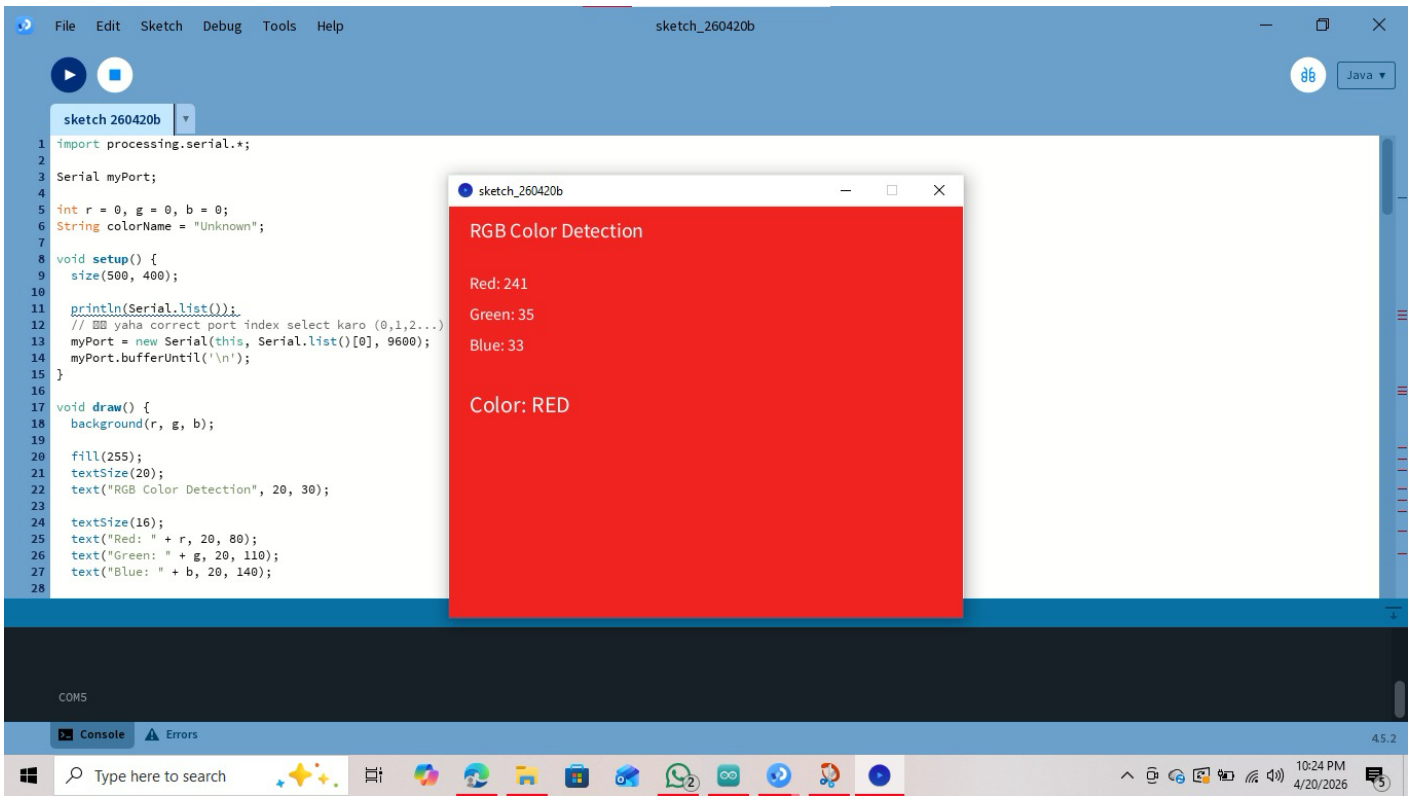


Figure 3 : Output on Processing App

Learning outcomes (What I have learnt):

1. Understood RGB color detection concept
2. Learned how to interface sensors with Arduino
3. Gained knowledge of serial communication
4. Learned real-time data visualization using Processing
5. Improved practical skills in hardware and software integration